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A Credit Access Index For Youth Entrepreneurs in Non-Metropolitan Brazil

Research Proposal – IARCO 2026

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Class/Grade/Year: 11th grade-2026
Country: Brazil
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Introduction

Access to credit is one of the central constraints on entrepreneurship in Brazil, where micro and small enterprises account for roughly a third of national GDP and generate the majority of formal jobs, yet have historically struggled to obtain financing from traditional banks [2]. Government-backed guarantee programs have expanded in recent years, but a 2024 Sebrae survey of small business owners found that only 45% of applicants secured the loan they sought, while 46% were denied outright, underscoring how far credit access still lags behind demand even under favorable federal programs [4].

This gap is not evenly distributed across the country. Digital lenders and peer-to-peer platforms have grown quickly in Brazil's metropolitan centers, but recent Central Bank research shows that fintech credit still concentrates on borrowers already partially served by banks, while traditional banks responded competitively only in markets with an existing fintech presence [1]. Academic work on fintech-driven financial inclusion has likewise focused on regional disparities between the North/Northwest and the wealthier South/Southwest, largely leaving sub-regional, town-level variation unexamined [2]. For young entrepreneurs in the interior of São Paulo state, small agribusiness and commercial towns removed from state national and financial hubs, this means there is currently no tool that identifies, at the municipal level, where credit access is weakest and why.

This proposal develops a Credit Access Index (CAI): a composite, town-level measure combining indicators of formal banking presence, fintech penetration, and informal lending prevalence, validated against the real loan experiences of youth entrepreneurs. The goal is a reusable diagnostic tool, rather than a one-off survey, that local development agencies such as Sebrae, municipal governments and youth business associations could use to target interventions toward the towns and populations where credit frictions are most severe.

Research Questions

This study aims to develop and validate the Credit Access Index (CAI) for non-metropolitan municipalities in the interior of São Paulo state. Specifically, the study intends to answer the following questions:

1. How effective is the CAI in capturing variation in credit accessibility across interior municipalities, in terms of: a. Formal banking presence (branch density, account penetration) b. Fintech and digital lending penetration c. Prevalence of informal or family-based lending d. Composite score reliability (internal consistency)
2. Is there a significant difference in CAI scores across municipalities of different population size and economic profile (e.g., agribusiness-cooperative towns vs. commerce-driven towns)?
3. To what extent does the CAI predict the real-world loan application outcomes reported by youth entrepreneurs (18–29) in the sampled municipalities, in terms of: a. Approval versus rejection b. Time to decision c. Interest rate or terms offered
4. What is the rate of acceptability of the CAI as a planning tool among local stakeholders (Sebrae representatives, municipal economic officers), in terms of: a. Functionality b. Clarity of the indicators c. Actionability for policy or program design

Literature Review

Composite financial inclusion indices are an established methodology internationally. Sarma's index for cross-country comparison combined banking penetration, availability, and usage into a single measure [6], and the approach has since been adapted subnationally: Mojica and Mapa built a provincial-level Financial Inclusion Index for the Philippines and found strong geographic clustering, with neighboring provinces showing similar inclusion levels [5]. The IMF has proposed a factor-analysis-weighted composite index to address earlier criticisms that index weights were chosen arbitrarily rather than statistically derived [7]. Comparable municipal-level work exists for Mexico, where researchers have applied Principal Component Analysis and Human-Development-Index-style aggregation to rank municipalities on financial inclusion [17]. None of this literature, however, has been applied at the town level within Brazil, nor tailored to the specific experience of youth entrepreneurs.

Within Brazil, research on fintech and small-business credit has focused mainly on macro or regional patterns. A Central Bank of Brazil working paper using near-complete data on unsecured working-capital loans found that peer-to-peer lenders tend to serve smaller and riskier firms already banked, and that their entry pushes incumbent banks to lower rates and extend credit to firms that previously struggled to borrow [1]. A thesis from Insper examined how fintech credit affects SME performance specifically in Brazil's North and Northeast, regions with lower bank branch density, and linked greater fintech presence to reduced unemployment in underserved areas [2]. Industry reporting corroborates that fintechs have taken on client segments, particularly smaller and newer firms, that large banks have been

retreating from [3], and that instant-payment infrastructure (Pix) and Open Finance data-sharing have accelerated financial inclusion nationally, even as gains remain concentrated in specific sectors and firm sizes [8].

Government-backed credit programs add another layer. Sebrae's Acredita/FAMPE guarantee fund has channeled billions of reais in guaranteed credit to small businesses since its expansion, and Sebrae's own leadership has publicly acknowledged persistent inequality in credit access as the motivation for the program [4], [9]. Even so, national survey data cited by Sebrae shows that historically only about 12% of small-business owners secured credit in a given period, and that formalization and credit access remain the two most commonly cited obstacles for small entrepreneurs generally [9]. Separately, Sebrae mapping work has identified millions of young Brazilians who begin entrepreneurial activity early, often in interior or non-metropolitan settings, but does not connect this population to town-level credit-access data [9].

Taken together, the literature supports two conclusions that motivate this study. First, composite indices are a well-validated way to summarize multidimensional credit or financial-inclusion conditions, but existing indices have not been built for Brazilian municipalities nor validated against individual entrepreneurs' loan outcomes. Second, Brazilian fintech and credit-access research has so far worked at the national or macro-regional level, leaving town-level disparities among interior municipalities (precisely where youth entrepreneurship is common but formal financial infrastructure is thin) unexamined. ForeSee-style acceptability testing with the people who would actually use a monitoring tool suggests a similar validation step is valuable here: measuring not just statistical performance, but whether local stakeholders find the resulting index usable [9].

Proposed Methodology

The method is a quantitative, index-construction and validation approach, structured in five stages.

1. Indicator Selection and Data Collection

Secondary data will be collected for a sample of 15–20 non-metropolitan municipalities in the interior of São Paulo state, varying in population size and economic base (e.g., agribusiness-cooperative towns versus commerce-driven towns). Indicators will be drawn from public sources including IBGE (population and economic census data), the Central Bank of Brazil (bank branch and correspondent density, credit-operation volume by municipality), and Sebrae publications (fintech and MEI registration data, guarantee-fund usage by region).

2. Index Construction

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Indicators will be normalized (min-max scaling to a 0–1 range) and aggregated using Principal Component Analysis to derive statistically-based weights, following the approach used in prior financial inclusion indices [7], [17], rather than assigning weights arbitrarily. The result is a single Credit Access Index (CAI) score per municipality, scaled 0–100.

3. Validation Survey

A structured survey will be distributed to youth entrepreneurs (ages 18–29) across the sampled municipalities, collecting self-reported loan application history: whether they applied for formal credit, whether the application was approved or rejected, time to decide, and terms offered. The survey will also ask about use of informal or family lending as an alternative.

4. Statistical Testing

The relationship between each municipality's CAI score and its entrepreneurs' reported loan outcomes will be tested using ordinal logistic regression (approval likelihood as the ordinal outcome, CAI and control variables as predictors). Differences in CAI scores across municipality clusters (by economic base) will be tested with the Kruskal-Wallis test, and categorical associations — for instance, between fintech usage and loan approval — will be tested with chi-square, consistent with the statistical approach used in the author's prior empirical research.

5. Acceptability Rating

Sebrae representatives and municipal economic-development officers will evaluate the CAI's functionality, clarity, and actionability using a 4-point Likert rubric, mirroring the acceptability-testing step used to validate applied monitoring tools with their intended end users.

6. Performance Evaluation and Recommendations

Overall CAI performance will be assessed through internal consistency (Cronbach's alpha across indicators), predictive validity against survey-reported loan outcomes, and stakeholder acceptability scores. Recommendations will focus on which indicators most explain interior credit gaps and how local agencies could prioritize interventions accordingly.

Expected Outcome

The study is expected to produce a validated, reusable Credit Access Index for non-metropolitan São Paulo municipalities, along with evidence on which components of the index, such as bank branch density, fintech penetration, or informal lending prevalence, most strongly predict real credit outcomes for youth entrepreneurs. Results will guide recommendations for Sebrae, municipal governments, or youth business associations seeking to target credit-access interventions toward the towns and populations facing the greatest frictions.

Potential Limitations

Because data collection and the validation survey are feasible for a solo student researcher within an 8-month window, the municipal sample is limited to 15–20 towns, which constrains statistical power and generalizability beyond the interior of São Paulo state. Self-reported loan outcomes may be subject to recall or social-desirability bias, and the choice of PCA weighting, while statistically grounded, still involves methodological judgment that could be revisited with a larger dataset in future work.

Conclusion

The Credit Access Index offers a practical, statistically grounded tool for identifying where credit frictions are most severe among youth entrepreneurs in Brazil's interior. By combining composite-index methodology with direct validation against entrepreneurs' real loan experiences and acceptability testing with the local agencies that would use it, this study aims to close a gap between international financial-inclusion measurement techniques and the specific, underexamined conditions facing non-metropolitan youth entrepreneurship in Brazil.

Project Practicalities

The project will span approximately 8 months. The first 2 months will involve indicator selection, secondary data collection, and survey instrument design. The next 4 months will be dedicated to distributing the validation survey, constructing the index, and running the statistical analyses. The final 2 months will focus on stakeholder acceptability testing, finalizing results, and preparing the full report.

Post-Program Plan

Upon completion, the researcher intends to refine the index with a larger municipal sample and share the findings with Sebrae's regional office and municipal economic-development contacts as a potential planning tool. The results will also be considered for submission to a relevant student economics publication or a future academic research competition.

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